Topic III: Johnson-Lindenstrauss

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We will now consider a famous randomized method of embedding data into low-dimensional space. We will also describe other uses of randomization in data science.

1 The Johnson-Lindenstrauss Lemma

Take data $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subset \mathbb{R}^p$. We seek a matrix $\mathbf{L} : \mathbb{R}^p \to \mathbb{R}^d$, for some $d \ll p$, that approximately preserves the pairwise distances between the data points, i.e. so that there is a small $\epsilon > 0$ such that

$$(1-\epsilon)\|\mathbf{x}_i - \mathbf{x}_j\| \le \|\mathbf{L}\mathbf{x}_i - \mathbf{L}\mathbf{x}_j\| \le (1+\epsilon)\|\mathbf{x}_i - \mathbf{x}_j\| \tag{1}$$

for all $\mathbf{x}_i$ and $\mathbf{x}_j$.

In general, it is not possible to perform dimensionality reduction without incurring some distortion in the distances. To see this, consider the following:

Proposition 1.1. *Let $\mathcal{X} = \{\mathbf{0}, \mathbf{e}_1, \dots, \mathbf{e}_p\} \subset \mathbb{R}^p$, where $\mathbf{e}_i$ denotes the i-th standard basis vector. If $f : \mathcal{X} \to \mathbb{R}^d$ and $\|f(\mathbf{x}) - f(\mathbf{y})\| = \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathcal{X}$, then $d \ge p$.*

Proof. By translating f (which does not change distances) we may assume that $f(\mathbf{0}) = \mathbf{0}$. Let $\mathbf{v}_i = f(\mathbf{e}_i)$, $1 \le i \le p$. Then $\|\mathbf{v}_i\| = 1$ for all $1 \le i \le p$, and for $i \ne j$, $\|\mathbf{v}_i - \mathbf{v}_j\| = \sqrt{2}$. Consequently,

$$2 = \|\mathbf{v}_i - \mathbf{v}_j\|^2 = \|\mathbf{v}_i\|^2 + \|\mathbf{v}_j\|^2 + \langle \mathbf{v}_i, \mathbf{v}_j \rangle = 2 + \langle \mathbf{v}_i, \mathbf{v}_j \rangle, \tag{2}$$

from which it follows that $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$. Therefore, $\mathbf{v}_1, \dots, \mathbf{v}_p$ are orthonormal, hence linearly independent; so $d \ge p$. □

We now explain the construction that produces a small distortion embedding with high probability. The analysis we will give is taken from [1]; the original Johnson-Lindenstrauss result may be found in [3].

Theorem 1.2 (Johnson-Lindenstrauss). *Suppose $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subset \mathbb{R}^p$. Let $\epsilon > 0$, and take an integer d satisfying*

$$d \ge \frac{2}{\epsilon^2} \log\left(n^2(n-1)\right). \tag{3}$$

Let $\mathbf{L} \in \mathbb{R}^{d \times p}$ be a random matrix with iid $\mathcal{N}(0, 1/d)$ entries. Then

$$\mathbb{P}\left\{(1-\epsilon) \le \frac{\|\mathbf{L}\mathbf{x}_i - \mathbf{L}\mathbf{x}_j\|}{\|\mathbf{x}_i - \mathbf{x}_j\|} \le (1+\epsilon)\, \forall \mathbf{x}_i \ne \mathbf{x}_j\right\} \ge 1 - \frac{1}{n}. \tag{4}$$

In other words, a *random* linear transformation into $O(\log(n)/\epsilon^2)$-dimensional space produces an ϵ-distortion embedding with probability at least $1 - 1/n$. In fact, it can be shown that no linear transformation can obtain a better embedding (in terms of the dependence of the embedding dimension on n and ϵ) in general [4].

The proof rests on the following lemma:

Lemma 1.3. *Let $\epsilon > 0$ and $\delta > 0$, and take an integer d satisfying*

$$d \ge \frac{2}{\epsilon^2} \log\left(\frac{2}{\delta}\right). \tag{5}$$

Let $\mathbf{L} \in \mathbb{R}^{d \times p}$ be a random matrix with entries that are iid $\mathcal{N}(0, 1/d)$. Then for any unit vector $\mathbf{v} \in \mathbb{R}^p$,

$$\mathbb{P}\left\{(1-\epsilon) \le \|\mathbf{L}\mathbf{v}\| \le (1+\epsilon)\right\} \ge 1 - \delta. \tag{6}$$

Before proving the lemma, let's see how Johnson-Lindenstrauss follows from it. Define

$$\delta = \frac{2}{n^2(n-1)}. \tag{7}$$

Then for all $\mathbf{x}_i \neq \mathbf{x}_j$, the lemma implies

$$\mathbb{P}\left\{(1-\epsilon) \le \frac{\|\mathbf{L}\mathbf{x}_i - \mathbf{L}\mathbf{x}_j\|}{\|\mathbf{x}_i - \mathbf{x}_j\|} \le (1+\epsilon)\right\} \ge 1-\delta, \tag{8}$$

and by the union bound,

$$\begin{aligned}
\mathbb{P}\left\{(1-\epsilon) \le \frac{\|\mathbf{L}\mathbf{x}_i - \mathbf{L}\mathbf{x}_j\|}{\|\mathbf{x}_i - \mathbf{x}_j\|} \le (1+\epsilon)\, \forall \mathbf{x}_i \neq \mathbf{x}_j\right\} &\ge 1 - \sum_{i<j} \mathbb{P}\left\{(1-\epsilon) \le \frac{\|\mathbf{L}\mathbf{x}_i - \mathbf{L}\mathbf{x}_j\|}{\|\mathbf{x}_i - \mathbf{x}_j\|} \le (1+\epsilon) \text{ fails}\right\} \\
&\ge 1 - \binom{n}{2}\delta \\
&= 1 - \frac{n(n-1)}{2}\frac{2}{n^2(n-1)} \\
&= 1 - \frac{1}{n},
\end{aligned} \tag{9}$$

which is the desired bound on the probability. We now prove Lemma 1.3.

Proof of Lemma 1.3. Let $\mathbf{R} = \sqrt{d} \cdot \mathbf{L}$, so $R_{ij} \sim \mathcal{N}(0,1)$. Then $X \equiv \mathbf{R}\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$. We will prove:

$$\mathbb{P}\left\{\|X\|^2 > (1+\epsilon)^2 d\right\} \le \frac{\delta}{2}, \tag{10}$$

and

$$\mathbb{P}\left\{\|X\|^2 < (1-\epsilon)^2 d\right\} \le \frac{\delta}{2}, \tag{11}$$

from which the union bound shows

$$\begin{aligned}
\mathbb{P}\left\{(1-\epsilon) \le \|\mathbf{L}\mathbf{v}\| \le (1+\epsilon)\right\} &= \mathbb{P}\left\{(1-\epsilon)^2 d \le \|\mathbf{R}\mathbf{v}\|^2 \le (1+\epsilon)^2 d\right\} \\
&= \mathbb{P}\left\{(1-\epsilon)^2 d \le \|X\|^2 \le (1+\epsilon)^2 d\right\} \\
&= 1 - \mathbb{P}\left\{\|X\|^2 < (1-\epsilon)^2 d \text{ or } \|X\|^2 > (1+\epsilon)^2 d\right\} \\
&= 1 - \left(\mathbb{P}\left\{\|X\|^2 < (1-\epsilon)^2 d\right\} + \mathbb{P}\left\{\|X\|^2 > (1+\epsilon)^2 d\right\}\right) \\
&\ge 1 - (\delta/2 + \delta/2) \\
&= 1 - \delta,
\end{aligned} \tag{12}$$

which is the desired bound and completes the proof.

To show (10), we recall that if $z \sim \mathcal{N}(0,1)$ and $b < 1/2$, then

$$\mathbb{E}\left[\exp\left\{bz^2\right\}\right] = \frac{1}{\sqrt{1-2b}}. \tag{13}$$

Fix λ between 0 and 1/2. Then by Markov's Inequality,

$$
\begin{aligned}
\mathbb{P}\left\{\|X\|^2 > (1+\epsilon)^2 d\right\} &= \mathbb{P}\left\{\sum_{i=1}^d X_i^2 > (1+\epsilon)^2 d\right\} \\
&= \mathbb{P}\left\{\exp\left(\lambda \sum_{i=1}^d X_i^2\right) > \exp\left(\lambda(1+\epsilon)^2 d\right)\right\} \\
&\le \exp\left(-\lambda(1+\epsilon)^2 d\right) \mathbb{E}\left[\exp\left(\lambda \sum_{i=1}^d X_i^2\right)\right] \\
&= \exp\left(-\lambda(1+\epsilon)^2 d\right) \prod_{i=1}^d \mathbb{E}\left[\exp\left(\lambda X_i^2\right)\right] \\
&= \exp\left(-\lambda(1+\epsilon)^2 d\right) \prod_{i=1}^d \frac{1}{\sqrt{1-2\lambda}} \\
&= \exp\left(-\lambda(1+\epsilon)^2 d\right) \frac{1}{(1-2\lambda)^{d/2}} \\
&= \exp\left\{-\lambda(1+\epsilon)^2 d - \frac{d}{2}\log(1-2\lambda)\right\}. \qquad (14)
\end{aligned}
$$

By taking

$$
\lambda = \frac{1}{2}\left(1 - \frac{1}{(1+\epsilon)^2}\right), \qquad (15)
$$

which lies between 0 and 1/2, and using $\log(1+\epsilon) < \epsilon$, the bound becomes:

$$
\begin{aligned}
\mathbb{P}\left\{\|X\|^2 > (1+\epsilon)^2 d\right\} &\le \exp\left\{-\lambda(1+\epsilon)^2 d - \frac{d}{2}\log(1-2\lambda)\right\} \\
&= \exp\left\{-\frac{1}{2}\left(1 - \frac{1}{(1+\epsilon)^2}\right)(1+\epsilon)^2 d - \frac{d}{2}\log\left(1 - \left(1 - \frac{1}{(1+\epsilon)^2}\right)\right)\right\} \\
&= \exp\left\{-\frac{d}{2}\left((1+\epsilon)^2 - 1\right) - \frac{d}{2}\log\left(\frac{1}{(1+\epsilon)^2}\right)\right\} \\
&= \exp\left\{-\frac{d}{2}\left(\epsilon^2 + 2\epsilon\right) + d\log(1+\epsilon)\right\} \\
&\le \exp\left\{-\frac{d}{2}\left(\epsilon^2 + 2\epsilon\right) + d\epsilon\right\} \\
&= \exp\left\{-\frac{d\epsilon^2}{2}\right\}. \qquad (16)
\end{aligned}
$$

Now using that $d \ge (2/\epsilon^2)\log(2/\delta)$, the bound becomes

$$
\begin{aligned}
\mathbb{P}\left\{\|X\|^2 > (1+\epsilon)^2 d\right\} &\le \exp\left\{-\frac{d\epsilon^2}{2}\right\} \\
&\le \exp\left\{-\frac{2}{\epsilon^2}\log\left(\frac{2}{\delta}\right)\frac{\epsilon^2}{2}\right\} \\
&= \frac{\delta}{2}, \qquad (17)
\end{aligned}
$$

which establishes (10). The proof of (11) is similar and is left as an exercise. □

Remark 1. Instead of defining the embedding by a matrix $\mathbf{L}$ with iid Gaussian entries, a similar result can be shown using a random projection matrix $\mathbf{L} \in \mathbb{R}^{d\times p}$ (i.e. where $\mathbf{L}$ has random orthonormal columns). That is, randomly projecting $\mathcal{X}$ into $O(\log(n)/\epsilon^2)$-dimensional space produces an ϵ-distortion embedding with high probability.

1.1 Randomized algorithms

The Johnson-Lindenstrauss result is notable for being an early and particularly simple example of a randomized algorithm. Randomized algorithms are now widely used in many areas of data science and scientific computing. An especially successful class of methods are found in numerical linear algebra.

To get a flavor of this kind of method, consider the task of computing the SVD of a real, rank r matrix $\mathbf{A}$ of size p-by-n. A randomized algorithm, first described in [5], proceeds as follows. First, $\mathbf{A}$ is applied to $r+k$ iid $\mathcal{N}(\mathbf{0}, \mathbf{I}_n)$ random vectors $\mathbf{z}_1, \dots, \mathbf{z}_{r+k}$, where k is a small constant, e.g. $k = 10$ or $k = 20$. Call the resulting vectors $\mathbf{b}_j = \mathbf{A}\mathbf{z}_j$. It can be shown that with high probability, the vectors $\mathbf{b}_1, \dots, \mathbf{b}_{r+k}$ span the entire column space of $\mathbf{A}$. Performing Gram-Schmidt on $\mathbf{b}_1, \dots, \mathbf{b}_{r+k}$ then yields an orthonormal basis $\boldsymbol{\omega}_1, \dots, \boldsymbol{\omega}_r$ for the column space of $\mathbf{A}$. (One must take some care with Gram-Schmidt to ensure numerical stability; but this is beyond the scope of the present discussion.)

Form the matrix $\boldsymbol{\Omega} = [\boldsymbol{\omega}_1, \dots, \boldsymbol{\omega}_r]$, which is of size p-by-r and has orthonormal columns; then $\boldsymbol{\Omega}\boldsymbol{\Omega}^T$ is the orthogonal projection onto the column space of $\mathbf{A}$. Consequently, $\boldsymbol{\Omega}\boldsymbol{\Omega}^T\mathbf{A} = \mathbf{A}$. Applying $\mathbf{A}^T$ to the r columns of $\mathbf{W}$, we form the product $\boldsymbol{\Omega}^T\mathbf{A}$; this is a matrix of size r-by-n. Compute its rank r SVD: $\boldsymbol{\Omega}^T\mathbf{A} = \mathbf{W}\boldsymbol{\Sigma}\mathbf{V}^T$. Then setting $\mathbf{U} = \boldsymbol{\Omega}\mathbf{W}$, the SVD of $\mathbf{A}$ is $\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$. Indeed, the matrix $\mathbf{U} = \boldsymbol{\Omega}\mathbf{W}$ has orthogonal columns, because by the orthonormality of the columns of $\mathbf{W}$ and $\boldsymbol{\Omega}$,

$$\begin{aligned} \mathbf{U}^T\mathbf{U} &= \mathbf{W}^T\boldsymbol{\Omega}^T\boldsymbol{\Omega}\mathbf{W} \\ &= \mathbf{W}^T\mathbf{W} \\ &= \mathbf{I}_r; \end{aligned} \tag{18}$$

and since $\boldsymbol{\Omega}\boldsymbol{\Omega}^T\mathbf{A} = \mathbf{A}$, we have

$$\begin{aligned} \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T &= \boldsymbol{\Omega}\mathbf{W}\boldsymbol{\Sigma}\mathbf{V}^T \\ &= \boldsymbol{\Omega}\boldsymbol{\Omega}^T\mathbf{A} \\ &= \mathbf{A}. \end{aligned} \tag{19}$$

It can be shown, see e.g. [5, 2] that this algorithm succeeds with overwhelming probability (in most practical settings, the probability of failure will be less than the probability that the computer's hardware fails). This contrasts with Johnson-Lindenstrauss, which can fail with non-trivial probability.

Let us now examine the cost of the randomized SVD described herein. Let $C_{\mathbf{A}}$ and $C_{\mathbf{A}^T}$ denote the cost of applying, respectively, $\mathbf{A}$ and $\mathbf{A}^T$ to a vector. Then the randomized algorithm describe above requires applying $\mathbf{A}$ to $O(r)$ vectors, at a cost of $O(r \cdot C_{\mathbf{A}})$; performing Gram-Schmidt on $O(r)$ vectors in $\mathbb{R}^p$, at a cost of $O(r^2p)$; applying $\mathbf{A}^T$ to r vectors, at a cost of $O(r \cdot C_{\mathbf{A}^T})$; and computing an SVD of the r-by-n matrix $\mathbf{W}^T\mathbf{A}$, at a cost of $O(r^2n)$. The total cost, therefore, is

$$O(r \cdot C_{\mathbf{A}} + r^2p + r \cdot C_{\mathbf{A}^T} + r^2n). \tag{20}$$

Generically (that is, for a matrix $\mathbf{A}$ without any special structure such as sparsity), we will have $C_{\mathbf{A}} = O(pn)$ and $C_{\mathbf{A}^T} = (pn)$. The cost in this case will be $O(rpn)$, which matches the cost of using, for instance, the Lanczos method. However, the Lanczos method, being iterative, is more difficult to parallelize and can also be quite delicate numerically. By contrast, the randomized algorithm is more easily parallelizable and numerically stable, and hence is a more attractive method.

The scheme described must be modified if the matrix $\mathbf{A}$ is not rank r (but one wishes to only compute the top r components). Roughly speaking, the idea is to apply $\mathbf{A}$ and $\mathbf{A}^T$ iteratively, which increases the gap between r-th and $(r+1)$-st singular values of $\mathbf{A}$. Refer to Section 4.5 in [2] for details.

We also remark that there are faster versions of the randomized method described here that have cost $O(\log(r)pn)$. These can be obtained by replacing the Gaussian random vectors $\mathbf{z}_1, \dots, \mathbf{z}_{r+k}$ with a more structure distribution, e.g. obtained by randomizing and subsampling the discrete Fourier transform, that permit faster evaluation of the products $\mathbf{A}\mathbf{z}_j$. See Section 4.6 in [2] for details.

References

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